

a

Capability profile of network-inference

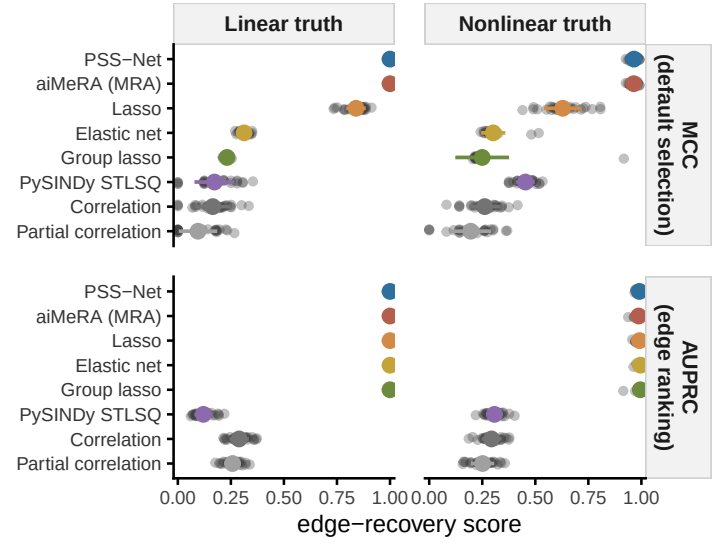
Perturbed steady-state Signed Nonlinear Partial Support scalar compositional
input (X*, edges edge functions edge function topology aware

	Yes	Yes	Yes	Yes	Yes	Yes	No
PSS-Net (ADSIHT)	Yes	Yes	Yes	Yes	Yes	Yes	No
MRA	Yes	Yes	Yes	No	No	Yes	No
Lasso / Elastic net	Yes	Yes	Yes	Partial	Yes	Yes	No
Group lasso	Yes	Yes	Yes	Partial	Yes	Yes	No
PySINDy STLSQ (PSS)	Yes	Yes	Yes	Yes	Yes	Yes	No
GENIE3 / GRNBoost2	No	Yes	No	Yes	Partial	Yes	Partial
GIES (interventional)	Yes	Yes	No	No	Yes	Partial	No
Correlation / partial cor.	No	No	Yes	No	No	Partial	No
SPIEC-EASI / SparCC	No	No	Partial	No	Partial	Partial	Yes

b

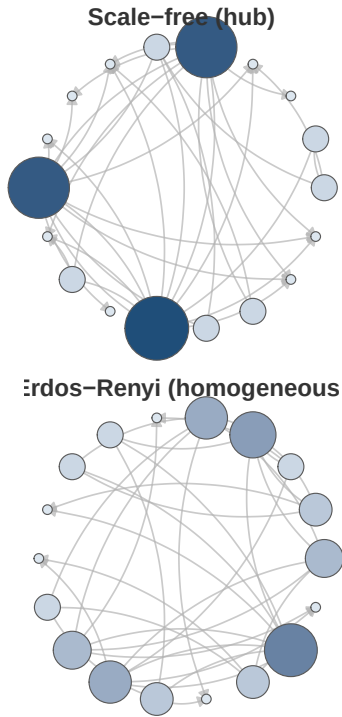
External benchmark separates edge selection

M = 2; N = 171 (N / [s log(p)] = 25.1). PSS-Net/MRA: |row-norm|



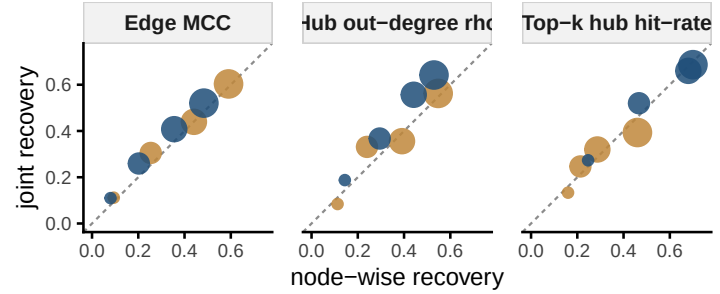
c

Two contrasting network topologies



d

Joint vs node-wise PSS-Net (above dashed line = joint)



Scale-free (hub) Erdos-Renyi (homogeneous) N 16 24 32

e

Compositional measurements degrade PSS network recovery

Dashed line marks oracle absolute-abundance MCC; points are simulation seeds

